SOFTWARE REQUIREMENTS SPECIFICATION

DeenWallet Transfer

Mobile money transfers, payments and services for Sierra Leone

Document	Software Requirements Specification (SRS)
System	DeenWallet Transfer: customer app, admin app, payment pages and server
Version	1.0
Date	5 November 2026
Status	Baseline of the system as built, with planned items marked
Product owner	Abdul Deen Kamara
Classification	Confidential. For the product owner, developers and authorised partners only.

This document follows the structure of IEEE 830 / ISO/IEC/IEEE 29148. Every requirement has an ID so it can be tested, changed and traced. Items that are built but switched off, or only planned, are marked clearly and are not claimed as working.

Document control

Version	Date	Author	Change
0.1	21 October 2026	Project team	First notes: accounts, conversion, bank transfer.
0.9	3 November 2026	Project team	Added payment links, card and crypto structure, admin sections.
1.0	5 November 2026	Project team	First complete SRS: all modules, interfaces, non-functional requirements, status matrix, open issues.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes what DeenWallet Transfer must do and the rules it must follow. It is the single reference for the product owner, developers, testers and partners (for example the payment providers). It states the requirements in a form that can be checked: each one has an ID, a priority and a status.

1.2 Scope

DeenWallet Transfer lets people in Sierra Leone move money between Orange Money and Africell Money, send money from mobile money to bank accounts, buy WAEC result pins, and collect payments through shareable payment links. A separate admin app lets the DeenWallet team run the service: review identity documents, handle failed payouts and refunds, manage staff access, read audit logs, answer customer email and watch the health and security of the server.

In scope: the customer app, the admin app, the public payment page, the server (API, scheduled jobs, database) and the operating tools that monitor and back up the server.

Out of scope: the providers' own systems (Monime, HoneyCoin, mobile network operators, banks), the Android store or download hosting, and any service marked *Planned* or *Pending* until it is approved and built.

1.3 Definitions, acronyms and abbreviations

Term	Meaning
SLE	Sierra Leonean Leone, the only currency used for customer amounts. Money is stored in the smallest unit ("minor" units): 100 = SLE 1.00.
Orange Money, Africell Money	The two mobile money services supported (provider ids m17 and m18).
Monime	The payment provider used to collect money (payment codes / USSD, card checkout) and to pay out to mobile money and banks.
HoneyCoin	A provider for crypto purchases with mobile money. Built in the system but switched off.
KYC	Know Your Customer: identity verification with an ID document.
WAEC result pin	A pin and serial number a student uses to view exam results on the WAEC portal.
Payment link	A web address (pay.deenwallapp.com/code) that anyone can open to pay a person, without an account.
Payout	Sending money out to a mobile money number or a bank account.
Collection	Receiving money from a customer's mobile money number.
Customer / User	A person with a customer account in the customer app.
Admin, Super admin, Main super admin	Staff roles in the admin app (see 2.3).
Section	A part of the admin app (for example KYC, Failed Payouts, WAEC Pins) that a super admin can switch on or off for each admin.
PIN	A 4-digit number the customer chooses to confirm money actions.
OTP	One-time code sent by email (6 digits, valid 10 minutes).
JWT	A signed login token. Access token valid 5 minutes, refresh token valid 7 days.
Webhook	A message a provider sends to the server when something happens (for example a payment was made).
Capacitor	The tool that wraps the web pages into the Android apps.
RPO / RTO	Recovery point objective (how much data can be lost) and recovery time objective (how long recovery may take).

1.4 References

- Project notes: docs/PROJECT-STRUCTURE.md, docs/ENVIRONMENT.md, docs/OPERATIONS.md (in the project).
- Monime developer documentation (collections, payouts, checkout sessions, webhooks).
- HoneyCoin developer documentation (authentication, on-ramps, webhooks, supported countries).
- IEEE Std 830-1998 and ISO/IEC/IEEE 29148:2018 (structure of this document).

1.5 Conventions

Item	Meaning
Requirement IDs	FR-xxx-nn = functional requirement of module xxx. NFR-xxx-nn = non-functional requirement.
"shall"	A mandatory behaviour. Each can be tested.
Priority	Must = needed for the service to work or be safe. Should = important. Could = useful if time allows (MoSCoW).
Implemented	Built, deployed and in use.
Delivered	Built and handed over as an update; deployment still to be confirmed by the owner.
Built, off	Built but switched off by a setting until an outside condition is met (for example a provider approval).
Pending	Waiting for an outside answer before it can be built or used.
Planned / Proposed	Agreed or suggested for the future. Not built.
Targets	Numbers in section 5 (speed, availability, recovery) are proposed targets for the owner to confirm. They are not measurements.

1.6 Overview of this document

Section 2 describes the product and its users. Section 3 lists the functional requirements module by module. Section 4 covers interfaces to people, providers and other systems. Section 5 gives the non-functional requirements (security, performance, reliability). Section 6 describes the data. Section 7 contains the models and diagrams. The appendices hold the status matrix, scheduled jobs, rate limits, acceptance scenarios, risks and open issues.

2. Overall description

2.1 Product perspective

DeenWallet Transfer is a new, self-contained product. It does not hold customer money in its own wallet. Money moves through the payment provider: the customer approves a payment from their own mobile money number, the provider reports it to the server, and the server instructs the provider to pay out to the destination. The server keeps the records, applies the rules and fees, and gives the staff tools to handle problems.

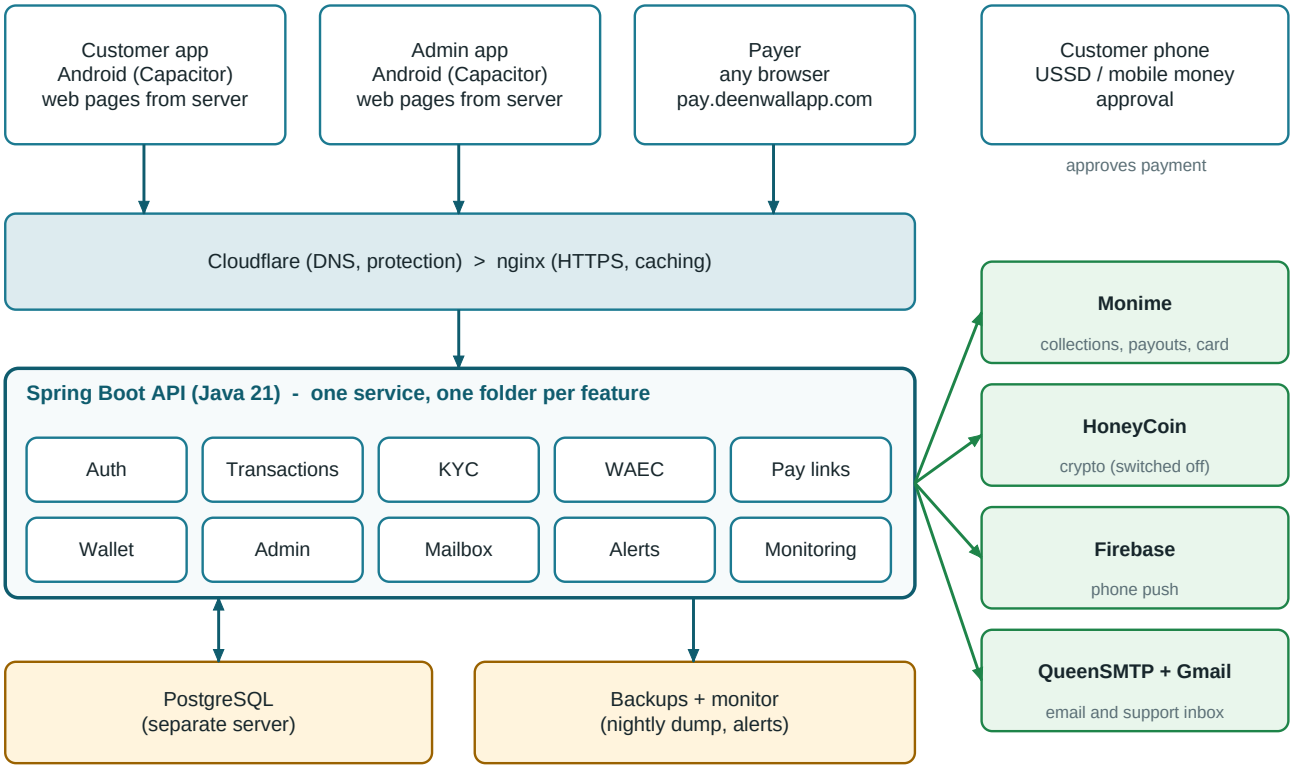


Figure 1 - System context. Arrows show the main calls. The customer approves each payment on their own phone.

2.2 Product functions (summary)

- **Accounts:** register with email verification, log in, 4-digit PIN, fingerprint login, trusted devices, password and PIN reset, profile photo, account deletion.
- **Convert money:** Orange Money to Africell Money and back, with a clear fee shown first.
- **Bank transfer:** from mobile money to a bank account, with account name verification and saved recipients.
- **KYC:** identity verification that raises daily limits.
- **WAEC result pins:** buy up to five pins, delivered in the app and by email.
- **Payment links:** create a link, share it, get paid straight to mobile money or a bank account; payers need no account.
- **Card payments and crypto purchase:** built and switched off until the providers approve them.
- **Admin app:** users, KYC review, failed payouts and refunds, wallet, WAEC stock, email inbox, audit log, server metrics, security, payment links, crypto orders and staff access control.
- **Operations:** pop-up alerts, monitoring every minute, nightly database backups.

2.3 User classes and characteristics

User class	Description	Main needs
Customer	Mobile money user in Sierra Leone; may have low technical skill; uses an Android phone.	Fast, clear transfers; fees shown before paying; receipts; safety of money.
Guest payer	Anyone who opens a payment link. No account.	Pay by Orange Money or Africell Money (card when enabled) in a few taps.
Admin	DeenWallet staff with access only to the sections a super admin switched on.	Handle customer problems in their own area.
Super admin	Senior staff; all sections; can edit admin details.	Oversight, wallet, security, staff control.
Main super admin	The head account. Cannot be controlled by anyone else; creates super admins.	Full control of the platform and its staff.
Provider systems	Monime and HoneyCoin sending webhooks.	Valid, signed messages answered quickly.

2.4 Operating environment

- **Server:** Ubuntu 24.04 (ARM64) virtual machine on Oracle Cloud. The Java service listens on 127.0.0.1:8082 behind nginx. Cloudflare sits in front.
- **Database:** PostgreSQL on a separate virtual machine in the same private network.
- **Runtime:** Java 21, Spring Boot 4.1, Maven build. Database changes run automatically from numbered migration files (V1 to V55).
- **Clients:** two Android apps (customer, admin) that load the web pages from the server; payment pages open in any modern browser.
- **Addresses:** api.deenwallapp.com (API and app pages), deenwallapp.com, pay.deenwallapp.com (payment links).

2.5 Design and implementation constraints

- Amounts are whole numbers of minor units; no floating point money.
- Customer amounts are in SLE only. Mobile money numbers are Sierra Leone numbers (+232, 8 digits).
- Only Orange Money (m17) and Africell Money (m18) are supported for mobile money.
- Provider limits apply (for example HoneyCoin lists SLE 5 to 15,000 for mobile money collections and payouts).
- No card number may ever pass through the DeenWallet server: card payment uses the provider's hosted checkout page.
- The app pages are served from fixed addresses that the Android apps, emails and nginx rely on; they must not be moved.
- Secrets (keys, passwords, tokens) live only in the server settings file, never in code.

2.6 Assumptions and dependencies

- Monime stays available and keeps supporting Orange Money and Africell Money in Sierra Leone.
- Card payments depend on Monime activating cards once the business registration is complete.
- The crypto service depends on HoneyCoin confirming Sierra Leone for the on-ramp, its fees and who performs customer ID checks.
- Crypto currency has no legal status in Sierra Leone (a grey area). Legal advice is needed before opening it to customers.
- Customers have a working email address and a phone that can approve mobile money requests.

3. System features (functional requirements)

Each feature has a short description followed by its requirements. "The system" means the DeenWallet server together with the apps and pages it serves.

3.1 Accounts and authentication (AUTH)

Lets a person create an account, prove who they are, and keep the account safe.

ID	Requirement	Priority	Status
FR-AUTH-01	The system shall let a person register with name, email, phone, password and a 4-digit PIN.	Must	**Implemented**
FR-AUTH-02	The system shall email a 6-digit code (valid 10 minutes) to confirm the email address before the account is fully active.	Must	**Implemented**
FR-AUTH-03	The system shall issue a short access token (5 minutes) and a refresh token (7 days) at login, and shall allow a refresh token to be revoked.	Must	**Implemented**
FR-AUTH-04	The system shall require the PIN (or the fingerprint) to confirm every money action. The PIN is checked on the server and is never sent with a transfer request.	Must	**Implemented**
FR-AUTH-05	The system shall temporarily lock an account after repeated wrong passwords or PINs, and shall also lock an IP address after 10 failed logins for 60 minutes.	Must	**Implemented**
FR-AUTH-06	The system shall offer fingerprint login using a device-bound credential after the customer turns it on.	Should	**Implemented**
FR-AUTH-07	The system shall list the devices signed in to an account ("Trusted devices") and let the customer remove them.	Should	**Implemented**
FR-AUTH-08	The system shall let a customer reset a forgotten password and a forgotten PIN using an emailed code.	Must	**Implemented**
FR-AUTH-09	The system shall let a customer set or remove a profile photo.	Could	**Implemented**
FR-AUTH-10	The system shall let a customer delete their account, and shall block deletion while money actions are still in progress.	Should	**Implemented**
FR-AUTH-11	The system shall record when a customer was last seen so staff can see who is online.	Could	**Implemented**

3.2 Mobile money conversion (MM)

The customer pays from one mobile money number and the money arrives on another number of the other network (Orange Money to Africell Money, or the reverse).

ID	Requirement	Priority	Status
FR-MM-01	The system shall show the amount, the fee and the total before the customer confirms (fee preview).	Must	**Implemented**
FR-MM-02	The system shall accept an amount of at least SLE 1.00 with at most 2 decimals, and at most SLE 50,000.00 per request, within the customer's daily KYC limit (a day is the calendar day in Freetown time).	Must	**Implemented**
FR-MM-03	The system shall detect the provider of a mobile money number and show the account holder name when available, so the customer can check the destination.	Must	**Implemented**

ID	Requirement	Priority	Status
FR-MM-04	The system shall create a payment code with Monime and give the customer the USSD steps to approve the payment on their own phone.	Must	**Implemented**
FR-MM-05	The system shall treat a transaction as paid only after the provider confirms it (webhook or direct status check).	Must	**Implemented**
FR-MM-06	The system shall, after payment is confirmed, pay out to the destination number and mark the transaction COMPLETED when the provider confirms the payout.	Must	**Implemented**
FR-MM-07	The system shall cancel an unpaid transaction after its payment code expires and shall never pay out an unpaid transaction.	Must	**Implemented**
FR-MM-08	The system shall block a duplicate request (same user, amount and number within a short window) and shall use idempotent provider references so one request can never be paid out twice.	Must	**Implemented**
FR-MM-09	The system shall show a receipt for every transaction and keep a history the customer can open.	Must	**Implemented**
FR-MM-10	If a payout fails after the customer paid, the system shall move the transaction to PAYOUT_FAILED, tell the customer their money is safe, and alert the admins (see 3.13).	Must	**Implemented**

3.3 Bank transfer (BK)

The customer pays from mobile money and the money is paid out to a bank account.

ID	Requirement	Priority	Status
FR-BK-01	The system shall show the list of supported banks with their logos.	Must	**Implemented**
FR-BK-02	The system shall verify the bank account and show the account holder name before the customer pays.	Must	**Implemented**
FR-BK-03	The system shall apply the same payment confirmation, payout, failure and refund rules as mobile money transfers.	Must	**Implemented**
FR-BK-04	The system shall apply the bank daily limit of the customer's KYC level.	Must	**Implemented**

3.4 Fees (FEE)

ID	Requirement	Priority	Status
FR-FEE-01	The system shall charge a 3% fee on conversion and bank transfers, shown to the customer as one line. Internally the fee is split into three equal 1% parts: provider deposit fee, DeenWallet revenue, provider withdrawal fee.	Must	**Implemented**
FR-FEE-02	The system shall calculate fees in whole minor units with half-up rounding and shall store the split on the transaction for reconciliation.	Must	**Implemented**
FR-FEE-03	For payment links, the payer shall pay a 3% service fee on top, so the link owner receives the full amount requested.	Must	**Implemented**
FR-FEE-04	For WAEC pins the system shall add a 2% fee to the pin price.	Must	**Implemented**
FR-FEE-05	Super admin withdrawals from the wallet shall keep a 1% service fee; admin withdrawals from their own wallet are free.	Should	**Implemented**

3.5 Saved recipients (REC)

ID	Requirement	Priority	Status
FR-REC-01	The system shall let a customer save mobile money and bank recipients and reuse them.	Should	**Implemented**
FR-REC-02	The system shall let a customer edit and delete saved recipients.	Should	**Implemented**

3.6 Identity verification and limits (KYC)

Customers who verify their email have basic limits. After an admin verifies an ID document, limits are much higher.

Level	Mobile money per day	Bank transfer per day	WAEC pins
Email verified (basic)	SLE 1,000	SLE 500	Allowed
ID verified	SLE 100,000	SLE 100,000	Allowed

ID	Requirement	Priority	Status
FR-KYC-01	The system shall let a customer submit a National ID card, a Driver's licence or a Passport (with the sides each document needs) and a selfie.	Must	**Implemented**
FR-KYC-02	The system shall store KYC documents encrypted and show them only to admins who have the KYC section.	Must	**Implemented**
FR-KYC-03	The system shall let an admin approve or reject a submission with a reason, and shall tell the customer the result.	Must	**Implemented**
FR-KYC-04	The system shall enforce the daily limit of the customer's level on every conversion and bank transfer and shall explain the limit when it is reached.	Must	**Implemented**
FR-KYC-05	The system shall alert the admins who have the KYC section when a new submission arrives.	Should	**Implemented**
FR-KYC-06	The limits shall be changeable by settings without changing code (KYC_BASIC_MOBILE_MONEY_DAILY_MINOR, KYC_BASIC_BANK_DAILY_MINOR, KYC_VERIFIED_DAILY_MINOR).	Should	**Implemented**

3.7 WAEC result pins (WAEC)

ID	Requirement	Priority	Status
FR-WAEC-01	The system shall let a customer buy 1 to 5 WAEC result pins (price per pin SLE 35.00 plus a 2% fee by default), paying from mobile money.	Must	**Implemented**
FR-WAEC-02	The system shall reserve pins for the order, release the reservation if unpaid after 30 minutes, and mark pins SOLD only after payment is confirmed.	Must	**Implemented**
FR-WAEC-03	The system shall store pins encrypted and shall show a pin to its buyer only, inside the app notification. A phone lock-screen alert shall never contain a pin.	Must	**Implemented**
FR-WAEC-04	The system shall email the pins to the address given and shall retry a failed email up to 5 times, then wait for an admin to resend to a corrected address.	Must	**Implemented**
FR-WAEC-05	The system shall alert the admins who have the WAEC Pins section when a pin email could not be delivered (first failure and when retries run out) and when stock runs out for a paid order.	Must	**Delivered**
FR-WAEC-06	The system shall let an admin upload pin stock, see stock and orders, and resend a pin email.	Must	**Implemented**

3.8 Payment links (PL)

A customer creates a link, shares it, and anyone can pay it from the browser. The money goes straight to the owner's mobile money number or bank account.

ID	Requirement	Priority	Status
FR-PL-01	The system shall let a customer create a link with a title, an optional note, a fixed amount or "payer chooses", one payment or many payments, an expiry (1, 7 or 30 days) and a destination (mobile money or bank).	Must	**Implemented**
FR-PL-02	The system shall give each link a short random code and the address https://pay.deenwallapp.com/code.	Must	**Implemented**
FR-PL-03	The payer shall need no account. They choose Orange Money or Africell Money, give their number, approve the payment on their phone, and see the result on the same page.	Must	**Implemented**
FR-PL-04	Each payment shall be a normal transaction, so confirmation, payout, failure handling, refunds and admin tools apply unchanged.	Must	**Implemented**
FR-PL-05	The system shall limit one guest payment to SLE 2,000 and one payer number to SLE 5,000 per 24 hours by default (settings APP_PAYLINK_GUEST_MAX_MINOR and APP_PAYLINK_GUEST_DAILY_MINOR).	Must	**Implemented**
FR-PL-06	The system shall limit a customer to 20 active links and shall let the owner close a link and see its payments.	Should	**Implemented**
FR-PL-07	The system shall give admins a Payment Links section to search any link, see its payments and close an abused link, and shall tell the owner when it is closed.	Should	**Implemented**
FR-PL-08	The link owner shall be notified when a payment arrives and when a payout problem delays it.	Should	**Implemented**

3.9 Card payments (CARD)

Card payment is built to use Monime's hosted checkout page. It is off by default (setting CARD_PAYMENTS_ENABLED=false). On the payment link page the card option shows "Coming soon".

ID	Requirement	Priority	Status
FR-CARD-01	The system shall create a Monime checkout session and redirect the payer to the provider's page. Card data shall never reach the DeenWallet server.	Must	**Built, off**
FR-CARD-02	The system shall mark a card payment paid only after the provider reports the checkout session completed, then follow the normal payout rules.	Must	**Built, off**
FR-CARD-03	If the provider rejects the session, the system shall remove the transaction and show a clear message that card payment is not available right now.	Must	**Built, off**
FR-CARD-04	Card refunds shall be done by an admin in the provider's dashboard, because the refund path of the system pays to a mobile money number.	Must	**Built, off**
FR-CARD-05	Switch-on condition: the business registration is completed and Monime activates card acceptance; then one small real payment shall be tested.	Must	**Pending**

3.10 Send to a crypto wallet (CRY)

The customer pays with mobile money and the provider sends USDT to the wallet address the customer gives (for example Binance). DeenWallet never holds the money or the crypto. Switched off by default (CRYPTO_ENABLED=false) and needs provider credentials.

ID	Requirement	Priority	Status
FR-CRY-01	The system shall show the feature as "Coming soon" unless crypto is enabled and the provider keys are set.	Must	**Built, off**
FR-CRY-02	The system shall allow only ID-verified customers (setting CRYPTO_REQUIRE_KYC), charge only the customer's own registered number, and ask for the PIN.	Must	**Built, off**
FR-CRY-03	The system shall check the amount limits (default SLE 5 to 2,000 each, SLE 5,000 per day), the wallet address format and the network (default USDT on BNB Smart Chain), and allow at most 2 open orders.	Must	**Built, off**
FR-CRY-04	The system shall record success only after the provider confirms both the order status and the charge status. A webhook alone shall never mark an order paid.	Must	**Built, off**
FR-CRY-05	The system shall alert super admins when a success arrives after a failure, and close orders never confirmed (30 minutes if never sent, 6 hours if pending).	Should	**Built, off**
FR-CRY-06	The admin app shall have a Crypto Orders section to search orders and ask the provider for the latest status.	Should	**Delivered**
FR-CRY-07	Before opening to customers: the provider shall confirm Sierra Leone for the on-ramp, its fees and who does customer ID checks; legal advice shall be obtained; sandbox and one small real test shall pass.	Must	**Pending**

3.11 Notifications and alerts (NOT)

ID	Requirement	Priority	Status
FR-NOT-01	The system shall keep an in-app notification list for each customer and admin, with unread counts.	Must	**Implemented**
FR-NOT-02	The system shall send a phone push notification (Firebase) for important events when the phone is registered.	Should	**Implemented**
FR-NOT-03	The admin app shall show a pop-up, vibrate and update the bell when a new alert arrives while it is open.	Should	**Implemented**
FR-NOT-04	The system shall alert admins about: failed payouts, refunds needing review, new KYC submissions, new customer emails, undelivered WAEC pin emails and WAEC stock-out.	Must	**Delivered**
FR-NOT-05	Alerts shall be selective: an admin shall receive an alert only if the matching section is switched on for them (WAEC Pins, Failed Payouts, KYC Verification, Messaging). Super admins shall receive all alerts.	Must	**Delivered**
FR-NOT-06	Tapping an alert shall open the matching section.	Should	**Delivered**

3.12 Admin app: users, roles and access (ADM)

ID	Requirement	Priority	Status
FR-ADM-01	The system shall have three admin levels: main super admin (cannot be controlled), super admin, admin. Only the main super admin creates super admins and controls them.	Must	**Implemented**
FR-ADM-02	A super admin shall be able to switch sections on or off for each admin (for example KYC, Wallet, Failed Payouts, WAEC Pins, Messaging, Payment Links, Crypto Orders). A request to a switched-off section shall be refused by the server.	Must	**Implemented**
FR-ADM-03	Creating an admin shall include a PIN. Admins shall not edit their own details; super admins can.	Must	**Implemented**
FR-ADM-04	Admin sign-in shall support an admin passcode and passkeys (WebAuthn), and recovery by email code.	Should	**Implemented**
FR-ADM-05	The system shall let an admin search and view customers, activate or deactivate an account, and see transactions with their status and failure reason.	Must	**Implemented**
FR-ADM-06	The system shall let an admin look up a transaction by its code (support lookup) and open the related payment link or crypto order.	Should	**Implemented**
FR-ADM-07	The admin app shall offer an Android update download protected by admin credentials, a short-lived download token and lockout after failed attempts.	Should	**Implemented**
FR-ADM-08	A super admin shall be able to set the support WhatsApp number shown to customers.	Could	**Implemented**

3.13 Failed payouts and refunds (FP)

When the customer has paid but the payout failed, the money is held and a person decides.

ID	Requirement	Priority	Status
FR-FP-01	The system shall list transactions in PAYOUT_FAILED with the reason, and let an admin RESEND the payout or REFUND the customer.	Must	**Implemented**

ID	Requirement	Priority	Status
FR-FP-02	A refund shall be paid back to the paying number as a mobile money payout. If the provider rejects the refund the transaction becomes NEEDS_REVIEW for manual handling.	Must	**Implemented**
FR-FP-03	Every resend and refund shall be recorded with who did it and when.	Must	**Implemented**
FR-FP-04	The system shall alert the admins who have the Failed Payouts section when a payout fails, fails again after a resend, or a refund needs review.	Must	**Delivered**
FR-FP-05	The system shall periodically check payouts and refunds still in progress with the provider (reconciliation) so none stays stuck.	Must	**Implemented**

3.14 Wallet (WAL)

The admin Wallet screen. The wording shown to staff says "DeenWallet", except the "Monime reference" line.

ID	Requirement	Priority	Status
FR-WAL-01	A super admin shall see the real provider balance, every payment and payout, and can withdraw to mobile money or bank and transfer money to an admin's wallet.	Must	**Implemented**
FR-WAL-02	An admin shall see their own wallet and withdraw from it free of charge.	Should	**Implemented**
FR-WAL-03	Withdrawals shall respect the single and daily limits (settings APP_WALLET_MAX_SINGLE_MINOR and APP_WALLET_MAX_DAILY_MINOR) and shall be recorded.	Must	**Implemented**
FR-WAL-04	The system shall reconcile wallet payouts with the provider every 30 seconds until they are final.	Must	**Implemented**

3.15 Money safety (SAF)

ID	Requirement	Priority	Status
FR-SAF-01	A super admin shall be able to freeze the money flow. While frozen the system stops new transfers, payouts, refunds and withdrawals. Money already collected is kept and paid out after the freeze is lifted.	Must	**Implemented**
FR-SAF-02	Freezing and resuming shall be recorded in the audit log.	Must	**Implemented**

3.16 Admin messaging (MSG)

ID	Requirement	Priority	Status
FR-MSG-01	The admin app shall have a mailbox that receives customer emails sent to the support address and lets staff read, reply and send new emails as the support address.	Must	**Implemented**
FR-MSG-02	The system shall send bulk emails to customers with a daily sending limit, and shall alert the admins who have Messaging when a customer email arrives.	Should	**Implemented**
FR-MSG-03	The system shall send automatic emails (codes, receipts) through the automatic mail service, separate from the support mailbox.	Must	**Implemented**

3.17 Audit (AUD)

ID	Requirement	Priority	Status
FR-AUD-01	The system shall record what each admin does (who, what, when, and on which item) in plain wording, covering users, KYC, WAEC, wallet, messaging, links, security and access changes.	Must	**Implemented**
FR-AUD-02	Super admins shall be able to filter the audit log by category and admin.	Should	**Implemented**
FR-AUD-03	Audit records shall be mirrored regularly so they survive clean-up of the main tables.	Should	**Implemented**

3.18 Monitoring and security (MON)

ID	Requirement	Priority	Status
FR-MON-01	The admin Server Metrics tab shall show server, app and database health, and Oracle usage against free-tier limits.	Should	**Implemented**
FR-MON-02	The system shall detect scanning and attack patterns (for example SQL injection attempts and SSH guessing), list them, and let a super admin block or remove an IP address.	Should	**Implemented**
FR-MON-03	The system shall collect error logs from the server and the apps, group them, and let a super admin read and clear them.	Should	**Implemented**
FR-MON-04	A small agent shall be able to report the health of an additional server to the Metrics tab every minute.	Could	**Implemented**
FR-MON-05	The system shall rate-limit sensitive endpoints (Appendix C).	Must	**Implemented**

3.19 Data lifecycle (DAT)

ID	Requirement	Priority	Status
FR-DAT-01	The system shall compute daily statistics every night.	Should	**Implemented**
FR-DAT-02	The system shall move transactions older than 180 days to an archive table, and delete expired codes, tokens and old logs after the configured retention (default 30 days).	Should	**Implemented**
FR-DAT-03	The system shall remove personal data of a deleted account as the retention rules allow.	Should	**Implemented**

3.20 Operations: alerts and backups (OPS)

ID	Requirement	Priority	Status
FR-OPS-01	The server shall be checked every minute: API answer, payment-link site, application and web server services, database reachability, disk space, age of the last backup, and certificate expiry.	Must	**Implemented**
FR-OPS-02	The owner shall be told by email (and optionally Telegram) after two failed checks in a row, again every 3 hours while the problem lasts, and when it is fixed. A pause command shall silence alerts during planned work.	Must	**Implemented**
FR-OPS-03	The database shall be backed up every night at 03:15, each backup shall be verified after writing (readable, contains the users table, not unusually small), and 14 daily, 8 weekly and 6 monthly copies shall be kept.	Must	**Implemented**
FR-OPS-04	A failed backup shall raise an alert and never replace the last good backup.	Must	**Implemented**
FR-OPS-05	Backups shall be copied to a place outside the application server.	Should	**Proposed**
FR-OPS-06	An outside watcher shall alert the owner if the whole server or the monitor goes silent.	Should	**Proposed**

3.21 Planned and pending features

These are not built. They are listed so the scope is clear and requirements can be written once approved.

ID	Feature	Notes	Status
FR-PLN-01	Bulk send	One payment for up to 20 recipients: the total plus 3% fee in a single payment, then one payout per recipient with a result for each.	Planned
FR-PLN-02	Subscriptions / recurring payments	Scheduled payments to a number or bank account. Mobile money needs the customer's approval on their phone each time unless the provider allows automatic charging.	Proposed
FR-PLN-03	Currency conversion tool	A rate calculator (SLE and USD). A real conversion service needs a provider.	Proposed
FR-PLN-04	Cross-border transfers	Depends on HoneyCoin enabling corridors for a Sierra Leone account. Their examples are KES/NGN to CNY. Not confirmed.	Pending
FR-PLN-05	USSD service flow	A full USSD menu of app features (code *715*2077842#). Blocked: the flow format and any call to our server are not documented by Monime.	Blocked
FR-PLN-06	Donations and group collections	Suggested next feature: one link, many contributors, a progress total.	Proposed
FR-PLN-07	Stuck-payment alert	Alert admins when a customer payment stays "waiting" too long.	Proposed
FR-PLN-08	Sender logo in Gmail (BIMI)	Deferred by the owner.	Proposed

4. External interface requirements

4.1 User interfaces

All screens are web pages shown inside the Android apps (and in a browser for payment links). They use the DeenWallet teal colour, large touch targets and short plain wording.

Interface	Page	Main screens
Customer app	index.html	Splash and logo, sign-in and register, home (swipe cards), Services, Convert money, Bank transfer, WAEC pins, Payment links, Send to crypto wallet (coming soon), Wallet history and receipts, Notifications, Support, Profile, KYC, Trusted devices, PIN and biometric settings, Terms, Privacy.
Admin app	admin.html	Sign-in (passcode, passkey), Home, Wallet, Audit, Messaging (email inbox), menu: Users, Admin management, KYC Verification, Failed Payouts, WAEC Pins, Payment Links, Crypto Orders, Server Metrics, Error Logs, Security, Profile.
Payment page	pay.html	Public page for one payment link: amount, provider choice (Orange Money, Africell Money, card "Coming soon"), number, progress and result.
Other pages	auth, welcome, transactions, reset pages, terms, privacy	Sign-in, first open, receipts, password and PIN reset opened from emails, legal text.

ID	Requirement	Priority	Status
NFR-UI-01	Every screen shall work on a phone screen of 360 px width without sideways scrolling.	Must	**Implemented**
NFR-UI-02	Every error message shall say in plain words what went wrong and what the person can do next.	Must	**Implemented**
NFR-UI-03	The apps shall open with the logo and go straight into the app, without a login flash for signed-in users.	Should	**Implemented**
NFR-UI-04	Money amounts shall always show as "SLE 1,234.56" and the fee and total shall be visible before confirmation.	Must	**Implemented**
NFR-UI-05	The back button of the phone shall go back one step inside the app and shall never bypass a security lock.	Should	**Implemented**

4.2 Hardware interfaces

The apps use the phone camera (KYC photos, profile photo), the fingerprint sensor (biometric login) and Android push services. No other special hardware is needed.

4.3 Software interfaces

System	Used for	How	Notes
Monime	Collect money (payment codes / USSD), payouts to mobile money and banks, bank list, card checkout sessions	HTTPS REST with an access token; webhooks signed with a secret	Production provider. Idempotent references. Card needs activation.
HoneyCoin	Crypto on-ramp (mobile money to USDT)	HTTPS REST with bearer token (1 hour); webhook with a shared secret header	Built, switched off. Status always re-checked directly.
Firebase Cloud Messaging	Phone push notifications	Server key file; web config endpoint	Optional per device.
QueenSMTP	Automatic email: codes, receipts, resets	SMTP/API with key	Sender no-reply address.
Gmail (IMAP and SMTP)	Support mailbox: receive and send as the support address	App password in the server settings	Daily send limit setting.
PostgreSQL	All data	JDBC to a separate server	Schema by numbered migrations.
Cloudflare	DNS, protection, real visitor IP	Proxy in front of nginx	Real IP restored in nginx.
nginx and certbot	HTTPS, caching, routing of three domains to the app	Reverse proxy to 127.0.0.1:8082	Certificates renewed automatically.
Oracle Cloud	Hosting and usage figures	VM, private network	Free-tier usage is tracked in the Metrics tab.

4.4 Communications interfaces

All traffic uses HTTPS. The API is JSON over HTTPS under /api/. Provider webhooks arrive at fixed public addresses and are verified before use.

Group	Examples of paths	Who
Auth	/api/auth/login, /register, /refresh, /forgot-password	Anyone (rate limited)
Customer	/api/transactions, /api/bank-transfers, /api/waec, /api/payment-requests, /api/crypto, /api/users/me	Signed-in customer
Public pay	/p/code, /api/public/pay/code (view, start, status)	Anyone with a link
Admin	/api/admin/... (users, kyc, waec, wallet, payment-links, crypto, audit, errors)	Admin or super admin, and only for allowed sections
Webhooks	/api/v1/webhooks/monime, /api/v1/webhooks/honeycoin	Providers (secret checked)
App updates	/api/app/android-update, /api/app/admin-android-update	Apps; admin download needs credentials
Monitoring	/api/monitor/agent/report	Extra-server agent (token)

5. Non-functional requirements

Numbers marked "target" are proposed values for the owner to confirm. They are not measured results.

5.1 Security (SEC)

ID	Requirement	Priority	Status
NFR-SEC-01	All communication shall use HTTPS. The server shall send security headers including a Content Security Policy.	Must	**Implemented**
NFR-SEC-02	Passwords and PINs shall be stored only as one-way hashes. Login tokens shall be signed (JWT) and short-lived (access 5 minutes).	Must	**Implemented**
NFR-SEC-03	KYC documents and WAEC pins shall be encrypted at rest with keys held only in the server settings.	Must	**Implemented**
NFR-SEC-04	Role checks shall be made on the server for every admin request; the admin API shall refuse a section the admin is denied, whatever the app shows.	Must	**Implemented**
NFR-SEC-05	The system shall rate-limit login, registration, codes, PIN checks, payments and lookups (Appendix C) and lock accounts and IPs after repeated failures.	Must	**Implemented**
NFR-SEC-06	Provider webhooks shall be authenticated (shared secret, constant-time comparison). A webhook shall only trigger a check; the result shall always be confirmed directly with the provider before money is moved or recorded.	Must	**Implemented**
NFR-SEC-07	Money actions shall be protected by PIN or fingerprint, and by the customer's own approval on their phone for collections.	Must	**Implemented**
NFR-SEC-08	Every admin action that changes data shall be written to the audit log.	Must	**Implemented**
NFR-SEC-09	Secrets shall not appear in code, logs, chat or screenshots. Settings files shall be readable only by the service owner.	Must	**Implemented**
NFR-SEC-10	Scanning and attack attempts shall be detected, shown to super admins and blockable.	Should	**Implemented**
NFR-SEC-11	The system shall not store card numbers and shall not ask for them (hosted provider page only).	Must	**Built, off**

5.2 Performance (PERF)

ID	Requirement	Priority	Status
NFR-PERF-01	Target: 95% of API requests that do not wait for a provider answer in under 1 second.	Should	**Proposed**
NFR-PERF-02	Web pages, scripts and images shall be compressed and cached by nginx so the apps open quickly; HTML shall revalidate so updates reach users.	Must	**Implemented**
NFR-PERF-03	A provider call shall never block the database: a payout shall not run inside one long database transaction.	Must	**Implemented**
NFR-PERF-04	Lookups (provider detection, account name) shall be cached or rate-limited so repeated taps do not overload the provider.	Should	**Implemented**
NFR-PERF-05	Target: a confirmed collection shall be paid out within 1 minute in normal conditions (provider time included).	Should	**Proposed**

5.3 Reliability and availability (REL)

ID	Requirement	Priority	Status
NFR-REL-01	A request that cannot be proven paid shall never be paid out; a request that may have been paid shall never be paid twice (idempotency, locking, reconciliation).	Must	**Implemented**
NFR-REL-02	Jobs shall retry and reconcile payouts, refunds, WAEC emails and wallet payouts until they are final or handed to a person.	Must	**Implemented**
NFR-REL-03	Target availability of the API: 99.5% per month.	Should	**Proposed**
NFR-REL-04	Before each deployment the working program file shall be kept (deenwallet.jar.prev) so a bad build can be reverted at once.	Must	**Implemented**
NFR-REL-05	The owner shall be alerted within about 3 minutes of the API stopping to answer.	Must	**Implemented**

5.4 Backup and recovery (BAK)

ID	Requirement	Priority	Status
NFR-BAK-01	Target RPO: at most 24 hours of data loss (nightly backup). Target RTO: restore within 4 hours.	Should	**Proposed**
NFR-BAK-02	Each backup shall be verified after writing and the newest shall be checked on demand.	Must	**Implemented**
NFR-BAK-03	A copy shall exist outside the application server.	Should	**Proposed**
NFR-BAK-04	A full restore shall be rehearsed on a spare database at least once before launch and then each quarter.	Should	**Proposed**

5.5 Usability and accessibility (USA)

ID	Requirement	Priority	Status
NFR-USA-01	A new customer shall be able to complete a first conversion without help text outside the app.	Should	**Proposed**
NFR-USA-02	Text shall be readable on small screens and in bright light (sufficient contrast), buttons at least 44 px high.	Should	**Implemented**
NFR-USA-03	Screens shall be usable on slow mobile data: lightweight pages, no heavy images.	Should	**Implemented**

5.6 Maintainability (MNT)

ID	Requirement	Priority	Status
NFR-MNT-01	Code shall be organised by feature (one folder per feature) and documented in docs/PROJECT-STRUCTURE.md.	Should	**Implemented**
NFR-MNT-02	Every setting shall be an environment variable with a safe default, documented in docs/ENVIRONMENT.md.	Should	**Implemented**
NFR-MNT-03	Database changes shall only be made by new numbered migrations. An applied migration shall never be edited.	Must	**Implemented**
NFR-MNT-04	Changes shall be delivered as tested update scripts that refuse to run on the wrong version of a file and do nothing when run twice.	Should	**Implemented**
NFR-MNT-05	The project shall have automated tests for money rules (fees, limits, state changes).	Should	**Proposed**

5.7 Legal and compliance (LEG)

ID	Requirement	Priority	Status
NFR-LEG-01	Terms of Use and Privacy Policy shall be shown in the app and kept up to date with identity verification.	Must	**Implemented**
NFR-LEG-02	Personal data shall be limited to what the service needs and removed according to the retention rules or on account deletion.	Must	**Implemented**
NFR-LEG-03	The business shall register and hold the licences Sierra Leone requires for payment services; the card service needs the registration.	Must	**Pending**
NFR-LEG-04	Legal advice shall be obtained before offering crypto purchases, because crypto has no legal status in Sierra Leone.	Must	**Pending**

6. Data requirements

Data is stored in PostgreSQL. The schema is built by 55 numbered migrations (V1 to V55). The main groups of data are below.

Group	Main records	Notes
People	users (role, PIN hash, lock fields, active flag, denied sections), trusted devices, biometric credentials, refresh tokens, email codes, password and PIN reset codes, profile photos	Admin levels and section switches live on the user record.
Money	transactions (type, status, amount, fee split, provider ids, failure reason, payer name, payment link id), payout recovery actions, transaction archive, payment requests (links), crypto orders	Amounts in minor units. Provider ids are unique to prevent double processing.
KYC	KYC submissions, document images (encrypted), audit of decisions	Approved submission means "verified".
WAEC	pins (encrypted, status AVAILABLE / RESERVED / SOLD), orders (delivery status and attempts), WAEC audit	
Wallet	admin wallets, withdrawals and fees, payout references, saved accounts	Reconciled with the provider.
Safety	money safety state (freeze) and limits	
Notifications	notifications, device tokens	Per user, including admins.
Admin and audit	audit log, admin passcodes, passkeys, PIN reset challenges	Audit is mirrored.
Mailbox	mailbox messages and sending safety counters	
Monitoring	error logs, security events, blocked and dismissed IPs, monthly usage, daily stats	Cleaned up by scheduled jobs.
Settings	app settings (support number)	Falls back to the environment default.

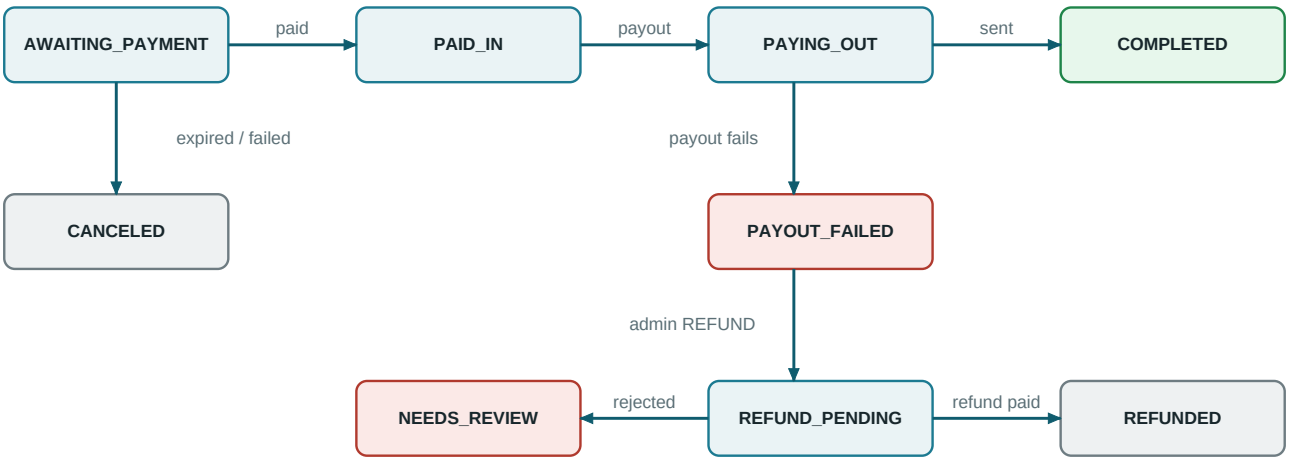
6.1 Data protection and retention

- KYC images and WAEC pins are encrypted; keys exist only in the server settings.
- Transactions older than 180 days move to an archive table; expired codes, tokens and old logs are removed after the retention period (default 30 days).
- A deleted account is cleaned as the rules allow. Deletion is blocked while money actions are in progress.
- Backups contain personal data and are readable by root only; any copy placed off the server shall be encrypted.

7. System models

7.1 Transaction lifecycle

Every money movement (mobile money, bank transfer, WAEC purchase, payment link payment) uses one set of states.



Only COMPLETED and REFUNDED are final for the customer. PAYOUT_FAILED and NEEDS_REVIEW always wait for a person.

Figure 2 - Transaction states. A person always resolves PAYOUT_FAILED and NEEDS_REVIEW.

State	Meaning
AWAITING_PAYMENT	Created. Waiting for the customer to approve the payment on their phone.
PAID_IN	The provider confirmed the money arrived. The payout has not started.
PAYING_OUT	The payout has been sent to the provider and is being confirmed.
COMPLETED	The destination received the money. Final.
FAILED / CANCELED	Not paid (expired or refused). Nothing was taken. Final.
PAYOUT_FAILED	Customer paid but the payout failed. Money is held. An admin resends or refunds.
REFUND_PENDING	A refund was approved and is being sent back to the paying number.
REFUNDED	The refund was paid. Final.
NEEDS_REVIEW	The refund was rejected by the provider. Handled by hand outside the app.

7.2 Main scenario: convert money

Step	Actor	Action
1	Customer	Enters the amount, source number and destination number. Sees the fee and the total.
2	Customer	Confirms with PIN or fingerprint.
3	System	Checks KYC limit, safety freeze and duplicates. Creates the transaction (AWAITING_PAYMENT) and a payment code at the provider.
4	Customer	Dials the USSD steps and approves on their own phone.
5	Provider	Sends a webhook. The system confirms the payment directly with the provider (PAID_IN).
6	System	Sends the payout to the destination (PAYING_OUT). On provider confirmation: COMPLETED, receipt and push notification.
7	System	If the payout fails: PAYOUT_FAILED, customer told their money is safe, admins with Failed Payouts alerted.

7.3 Main scenario: payment link

Step	Actor	Action
1	Owner	Creates a link (title, amount, type, expiry, destination) and shares it.
2	Payer	Opens pay.deenwallapp.com/code, picks Orange Money or Africell Money, enters their number.
3	System	Creates a normal transaction owned by the link owner; the payer pays the amount plus 3%.
4	Payer	Approves on their phone. The page shows progress and the result.
5	System	Pays out the full amount to the owner's mobile money or bank account. The owner is notified.

Appendix A. Implementation status by module

Module	Status	Notes
Accounts and authentication	**Implemented**	In use. Includes biometric, trusted devices, resets, photos, deletion.
Mobile money conversion	**Implemented**	In use. Real payouts confirmed in production.
Bank transfer	**Implemented**	In use.
KYC and limits	**Implemented**	In use.
WAEC result pins	**Implemented**	In use. Undelivered-email and stock-out alerts are Delivered (update to deploy).
Payment links (pay.deenwallapp.com)	**Implemented**	In use and confirmed working.
Card payments	**Built, off**	Built. Off until Monime activates cards after business registration.
Send to a crypto wallet	**Built, off**	Built (server, customer screen, admin section). Off. Waiting for HoneyCoin answers.
Admin app: users, roles, sections, wallet, audit, messaging	**Implemented**	In use.
Admin pop-up alerts for problems, selective by section	**Delivered**	Update delivered; deployment to be confirmed.
Money safety (freeze, limits)	**Implemented**	Deployed 1 November 2026.
Server metrics, security, error logs	**Implemented**	In use.
Operations: monitor and nightly backups	**Implemented**	Installed 4 November 2026. Test alert received; first backups verified.
Off-server backup copy, outside uptime watcher	**Proposed**	Recommended, not set up yet.
Bulk send, subscriptions, donations, cross-border, USSD flow	**Planned**	See 3.21.

Appendix B. Scheduled jobs

Job	When	What it does
Transaction expiry	Every minute	Cancels unpaid transactions after the payment code expires.
Refund scheduler	Every minute	Advances approved refunds.
Payout reconciliation	Every 2 minutes	Re-checks payouts still in progress with the provider.
Wallet reconcile	Every 30 seconds	Re-checks wallet payouts until final.
WAEC maintenance	10 minutes; 3 minutes	Releases stale pin reservations; retries pin emails.
Crypto reconcile	Every 2 minutes	Re-checks open crypto orders; closes stale ones.
Mailbox	30 s, 1 min, nightly	Fetches and sends mail, keeps the mailbox tidy.
Security scan, blocked IPs, usage	1 to 5 minutes	Detects attacks, applies blocks, records Oracle usage.
Server metrics	Every 5 seconds	Samples server health.
Money safety	Every 5 minutes	Checks safety state and limits.
Audit mirror	Every 5 minutes	Copies audit records to a safe table.
Nightly clean-up	03:15 to 03:45	Archive old transactions, daily statistics, retention, error log clean-up.
Server monitor (outside the app)	Every minute	Checks API, site, services, disk, certificates, backup age.
Database backup (outside the app)	Every night 03:15	Creates and verifies a backup, keeps 14 + 8 + 6 copies.

Appendix C. Rate limits

Counted per signed-in user where a user exists, otherwise per IP address. Exceeding a limit returns "too many requests".

Action	Endpoint	Max	Per
Login	`POST /api/auth/login`	10	5 min
Register	`POST /api/auth/register`	5	1 hour
Resend code	`POST /api/auth/resend-otp`	5	15 min
Forgot password / PIN	`POST /api/auth/forgot-...`	5	15 min
Confirm email	`POST /api/auth/confirm-email`	10	15 min
Reset password / PIN	`POST /api/auth/reset-...`	10	15 min
Refresh token	`POST /api/auth/refresh`	30	1 min
Biometric login	`POST /api/auth/biometric`	20	1 min
Admin recovery	`POST /api/admin/auth/recovery`	10	5 min
Admin APK authorise	`POST /api/app/admin-android-update/authorize`	5	5 min
Verify PIN	`POST /api/users/me/verify-pin`	10	5 min
Convert money	`POST /api/transactions`	6	1 min
Bank transfer	`POST /api/bank-transfers`	6	1 min
WAEC purchase	`POST /api/waec/purchases`	6	1 min
WAEC price quote	`GET /api/waec/quote`	60	1 min

Action	Endpoint	Max	Per
Provider detection	`GET /api/accounts/detect-provider`	20	1 min
Bank account check	`GET /api/bank-transfers/verify-account`	20	1 min
Fee preview	`GET /api/transactions/preview-fee`	60	1 min
Crypto order	`POST /api/crypto/orders`	6	1 min
Crypto quote	`GET /api/crypto/quote`	30	1 min

Appendix D. Acceptance scenarios

High-level checks that prove the main requirements. Each should be run on a test system or with small real amounts before a release that touches money.

ID	Scenario	Expected result	Covers
AT-01	Convert SLE 10 from Orange Money to Africell Money with the right PIN, then approve on the phone.	Transaction goes AWAITING_PAYMENT, PAID_IN, PAYING_OUT, COMPLETED. Receipt shows amount, fee and total. Destination receives SLE 10.	FR-MM-01..09
AT-02	Start a conversion and do not approve.	The transaction is cancelled after the code expires. Nothing is paid out.	FR-MM-07
AT-03	Send the same request twice quickly.	The second is refused as a duplicate. Only one payout is possible.	FR-MM-08
AT-04	Cause a payout failure after payment.	PAYOUT_FAILED. Customer sees "your money is safe". Admins with Failed Payouts get a pop-up; admins without it do not.	FR-MM-10, FR-FP-04, FR-NOT-05
AT-05	Admin refunds the failed payout.	REFUND_PENDING then REFUNDED; money returns to the paying number; the action is in the audit log.	FR-FP-01..03
AT-06	Basic customer converts above SLE 1,000 in a day.	Refused with a clear message about the limit. After KYC approval the higher limit applies.	FR-KYC-04
AT-07	Buy a WAEC pin with a wrong email address.	Pin is in the app. Email fails; admins with WAEC Pins get a pop-up; an admin can resend to a corrected address.	FR-WAEC-03..06
AT-08	Open a payment link on a phone browser and pay SLE 5.	Payer pays SLE 5 plus 3%. The owner receives SLE 5 and is notified.	FR-PL-02..08
AT-09	Switch the emergency freeze on, then try a transfer.	New transfers and withdrawals are refused. After resume, collected money is paid out.	FR-SAF-01
AT-10	Deny the KYC section for an admin and call the KYC API as that admin.	The server refuses the request.	FR-ADM-02, NFR-SEC-04
AT-11	Stop the application service for 3 minutes.	The owner gets a problem alert, then a recovery alert when it returns.	FR-OPS-01..02
AT-12	Run a backup, then check the newest file.	Backup file exists, is verified and the status screen shows its age under one hour.	FR-OPS-03..04
AT-13	Enter a wrong PIN repeatedly.	The account is locked for a short time with a clear message.	FR-AUTH-05
AT-14	With card payments off, open a payment link.	The card option shows "Coming soon" and cannot be chosen.	FR-CARD-01
AT-15	With crypto off, open the crypto screen.	It shows "Coming soon".	FR-CRY-01

Appendix E. Risks and mitigations

Risk	Impact	Mitigation
Provider (Monime) outage or delay	Payments and payouts stop or hang.	Reconciliation jobs, PAYOUT_FAILED handling, monitor and alerts, emergency freeze, status messages to customers.
Double payout	Direct money loss.	Idempotent references, locking, duplicate window, payout only after confirmed payment.
Server or database loss	Loss of records.	Nightly verified backups; add an off-server copy; rehearse a restore.
Server attack	Data theft, downtime.	Cloudflare, rate limits, attack detection and blocking, hashed secrets, encrypted documents, least-privilege sections.
Staff misuse	Wrong refunds or data exposure.	Section switches, audit log of every action, main super admin control, PIN on admin creation.
Regulation of payments in Sierra Leone	Service may need licences.	Register the business; confirm requirements with the regulator.
Crypto legal status	Legal and reputation risk.	Keep switched off; legal advice first; ID-verified users only; small limits.
Card chargebacks and fees	Cost and disputes.	Hosted provider page only; confirm fees; manual refunds in the provider dashboard.
Update delivered but not tested on the real build	Build or runtime error.	Build and test locally before deployment; keep deenwallet.jar.prev; test with small amounts.
Customer enters a wrong destination	Money to the wrong person.	Show the account holder name before paying; PIN confirmation; clear warnings.

Appendix F. Open issues

#	Issue	Owner action
1	The server still enforces a maximum of SLE 50,000.00 per request (in the transaction request checks and the fee preview), while an earlier note says the per-transaction cap was removed so that only the daily KYC limits apply.	Decide which is intended. If the cap should go, the limit in the request checks must be changed.
2	HoneyCoin: Sierra Leone for the crypto on-ramp, fees, markup and who does ID checks are unconfirmed. An email with these questions was sent.	Wait for the answer, then update FR-CRY and limits.
3	Monime card: not active until business registration; the card fee and settlement time are unknown.	Register the business; ask Monime for the fee and settlement.
4	Updates delivered and waiting for deployment: card button and card structure, crypto (server, customer screen, admin section), admin pop-up alerts, project tidy-up.	Apply in order, rebuild, test locally, then deploy.
5	The delivered Java was syntax-checked but not compiled against the full framework by the assistant; there are no automated tests for money rules yet.	Compile and test on the owner's machine; add tests (NFR-MNT-05).
6	Backups are only on the application server. The restore check reads the whole file; a restore into a spare database is not possible with the application's database user.	Set up an off-server copy; rehearse a restore on the database server.
7	No outside uptime watcher yet.	Create a free outside check and paste its address into the monitor settings.
8	USSD service flow is blocked by missing Monime documentation.	Ask Monime for the flow format and how a flow can call the server.
9	Business registration and any licences required for payment services.	Confirm with the regulator and a lawyer.
10	Targets in section 5 (speed, availability, RPO, RTO) are proposals.	Confirm or change them.

End of document.